A Practical Wrapper Protocol for Metadata-Hiding in Messaging

Lea Thiemt
FAU Erlangen-Nürnberg

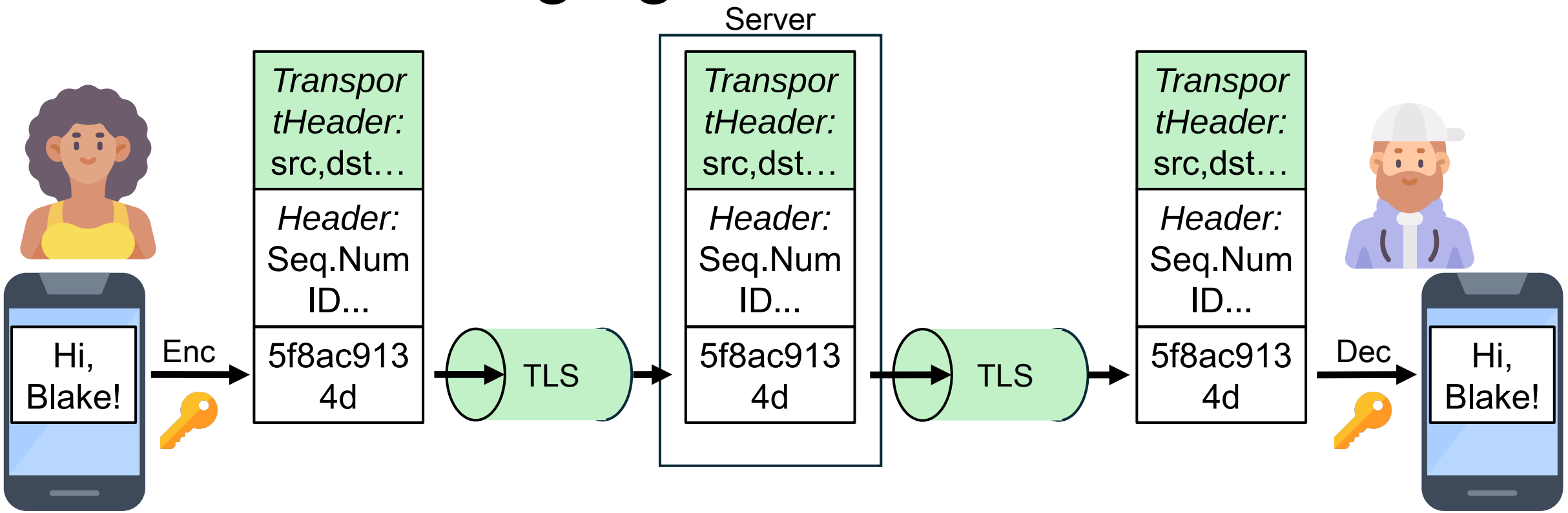
Paul Rösler
FAU Erlangen-Nürnberg

Alexander Bienstock
J.P. Morgan AI Research and J.P.
Morgan AlgoCRYPT CoE

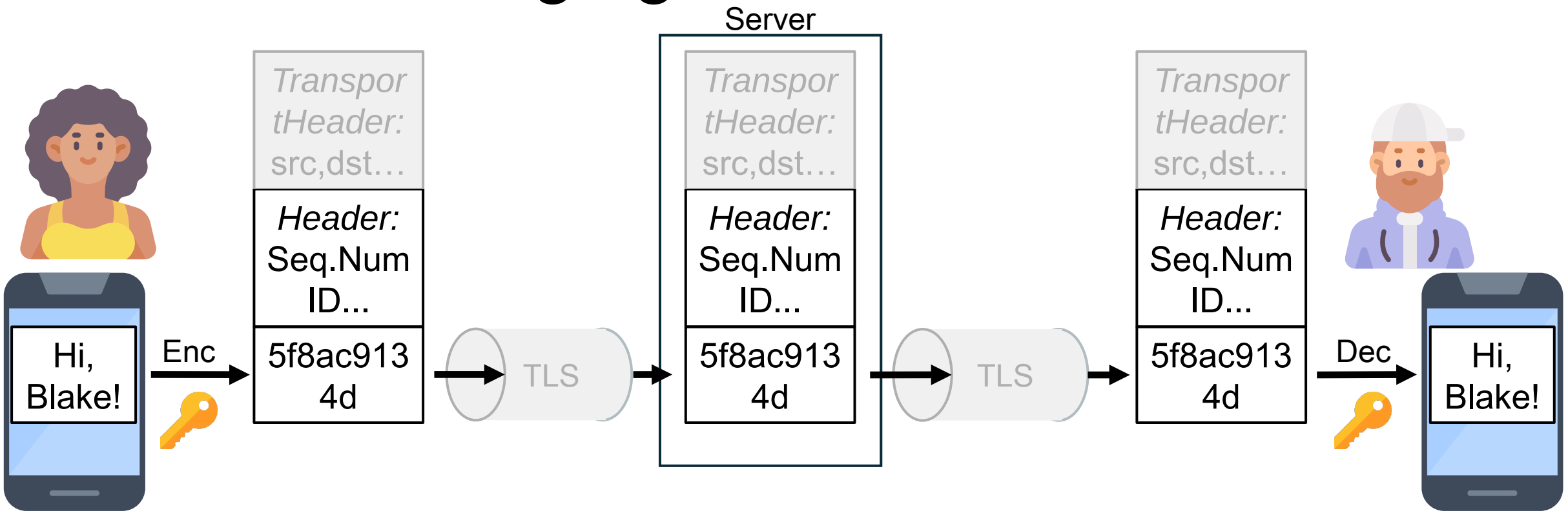
Rolfe Schmidt
Signal Messenger

Yevgeniy Dodis
New York University

Online Messaging

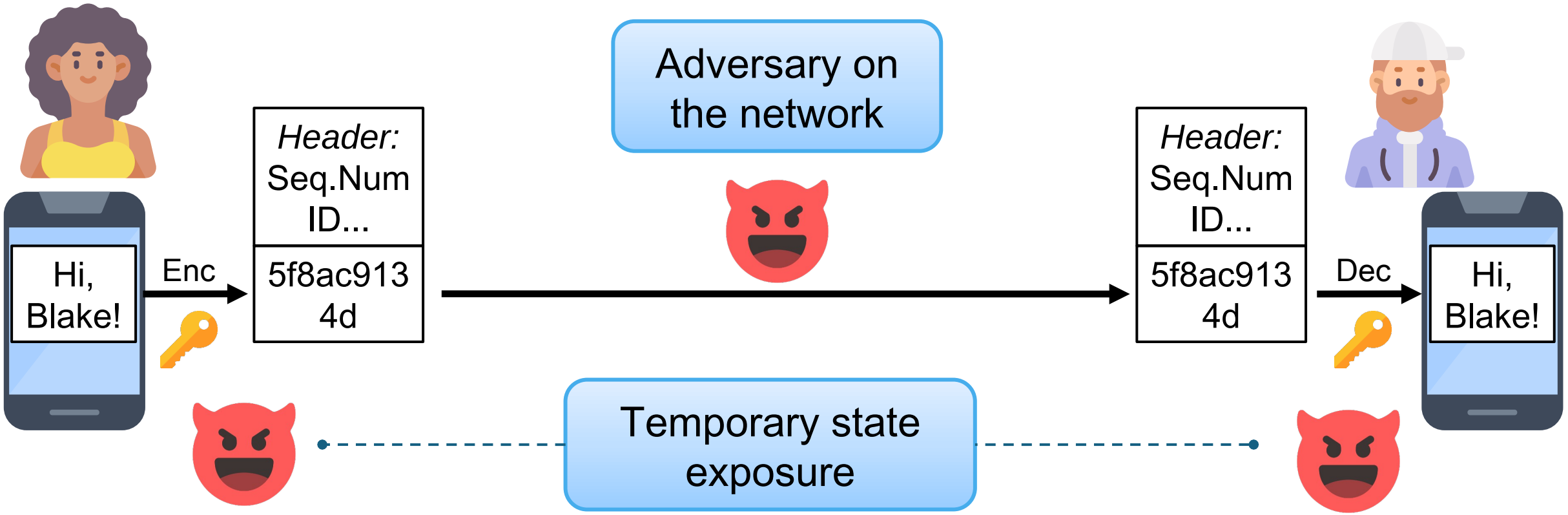


Online Messaging



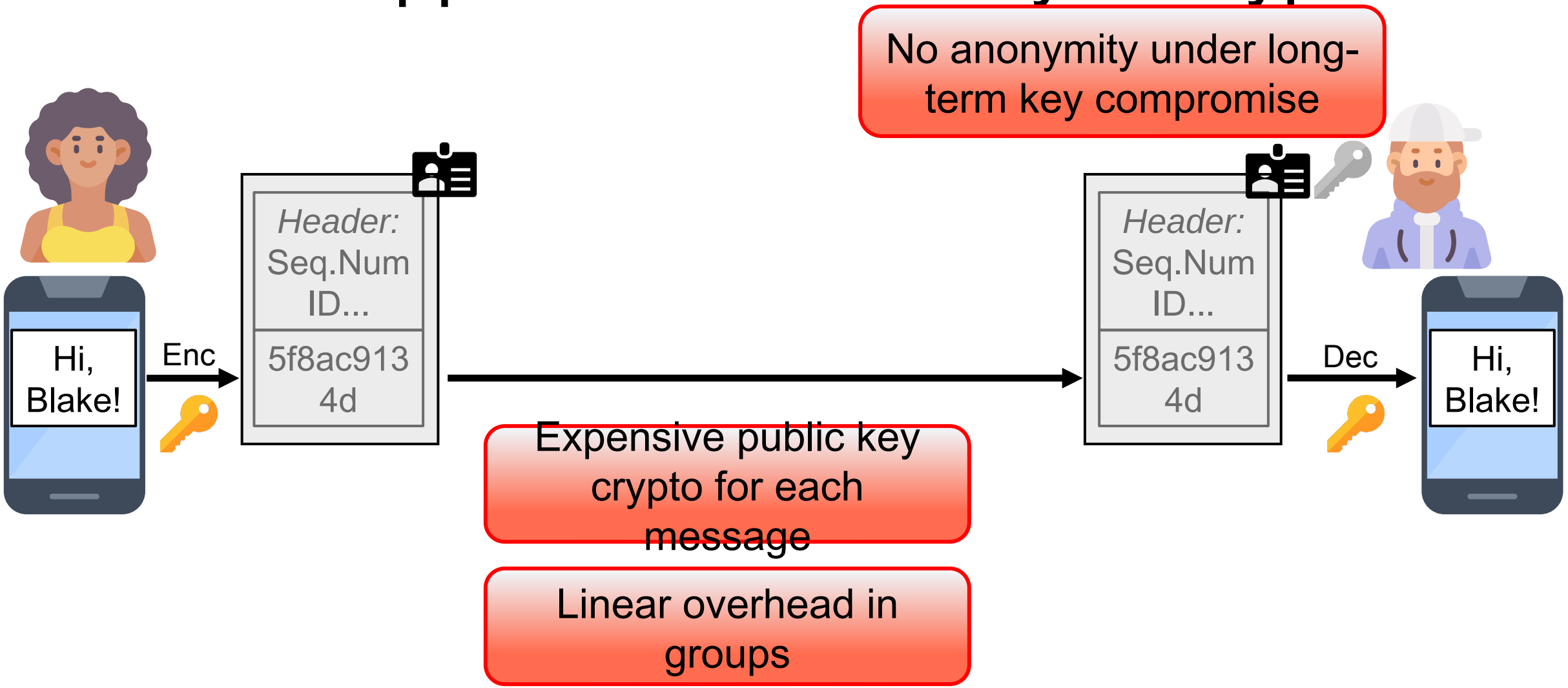
Goal: Hide Metadata on Ciphertext-Layer

Threat Model

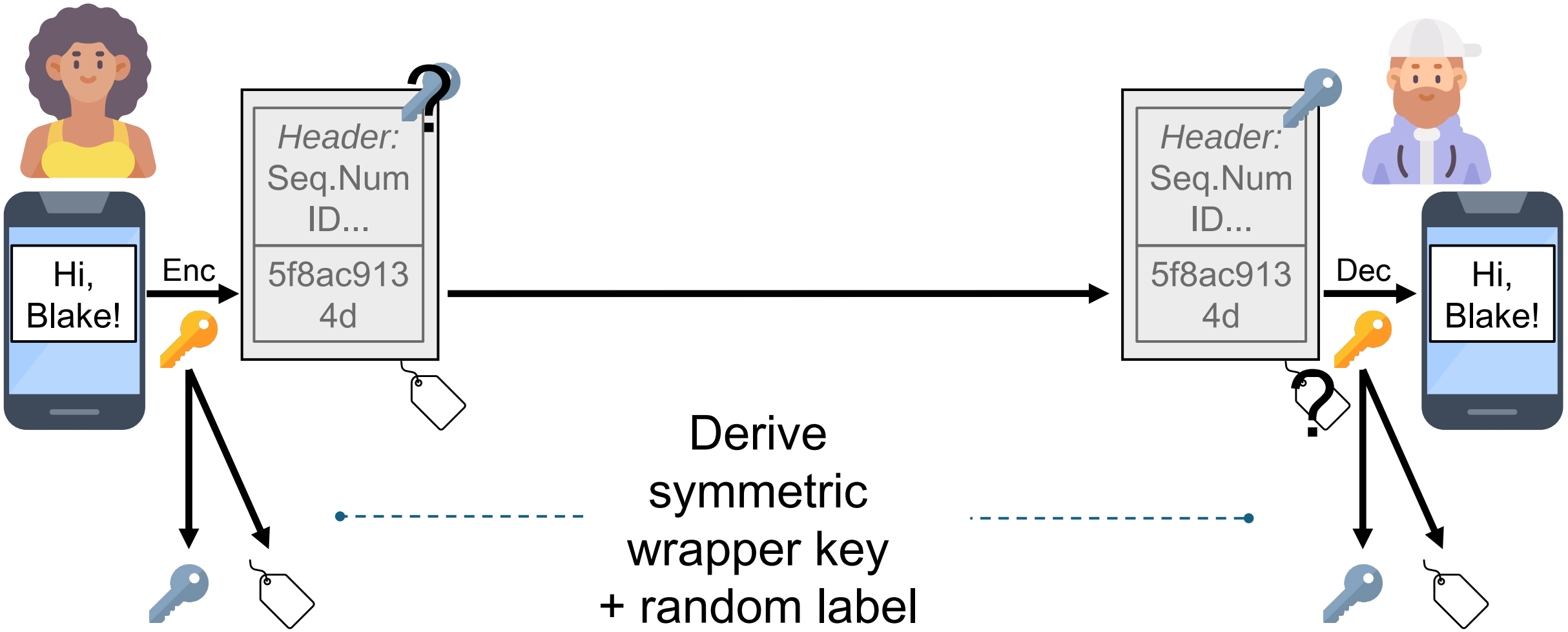


Goal: Hide Metadata on Ciphertext-Layer

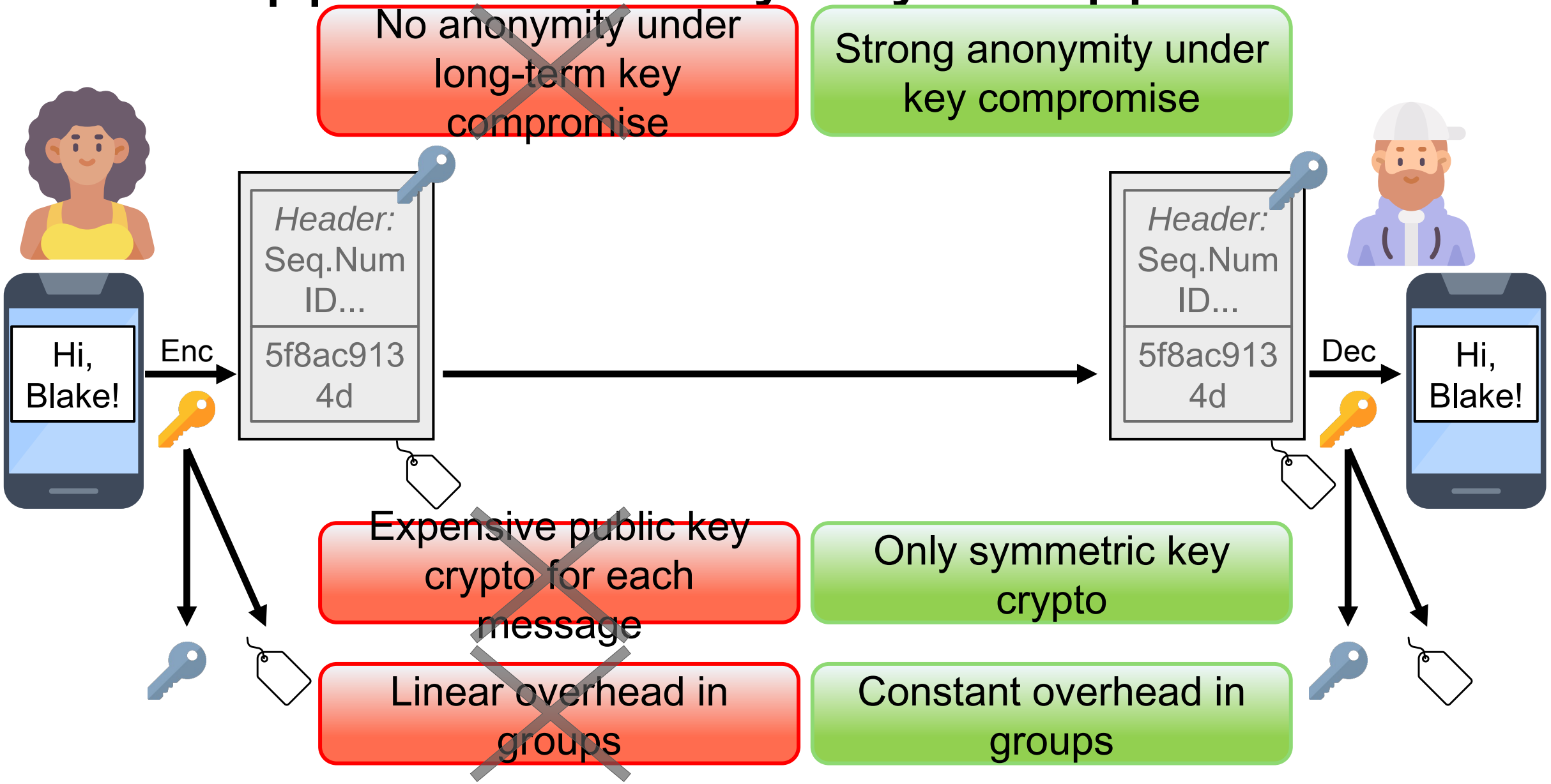
Current Approach: Public Key Encryption



Our Approach: Anonymity Wrapper



Our Approach: Anonymity Wrapper



Messaging 101

ck_0

$ck_1 \quad k_1$

$ck_2 \quad k_2$

$ck_3 \quad k_3$

$\cdots$

ck_0^*

$ck_1^* \quad k_1^*$

$c_1 = aenc_{k_1}(m_1)$

$c_2 = aenc_{k_2}(m_2)$

$c_3 = aenc_{k_3}(m_3)$

$\xrightarrow{c_1}$

$\xrightarrow{c_2}$

$\xrightarrow{c_3}$

$m_1 = adec_{k_1}(c_1)$

$m_2 = adec_{k_2}(c_2)$

$m_3 = adec_{k_3}(c_3)$

ck_0

$ck_1 \quad k_1$

$ck_2 \quad k_2$

$ck_3 \quad k_3$

$\cdots$

ck_0^*

$ck_1^* \quad k_1^*$

Metadata for Unreliable Networks

ck_0

$ck_1 \quad k_1$

$ck_2 \quad k_2$

$ck_3 \quad k_3$

$\cdots$

ck_0^*

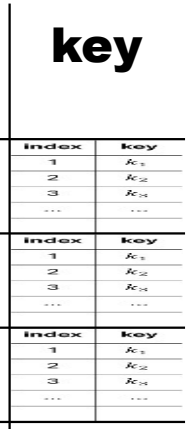
$ck_1^* \quad k_1^*$

$$c_1 = aenc_{k_1}(m_1)$$

$$c_2 = aenc_{k_2}(m_2)$$

$$c_3 = aenc_{k_3}(m_3)$$

c_3 + index=3 , ID →

index	key
1	
2	
3	
…	…

$$m_3 = aaec_{k_3}(c_3)$$

ck_0

$ck_1 \quad k_1$

$ck_2 \quad k_2$

$ck_3 \quad k_3$

$\cdots$

ck_0^*

$ck_1^* \quad k_1^*$

Anonymity Wrapper

ck_0

$ck_1 \quad k_1 \quad \boldsymbol{t_1}$

$ck_2 \quad k_2 \quad \boldsymbol{t_2}$

$ck_3 \quad k_3 \quad \boldsymbol{t_3}$

$\ldots$

ck_0^*

$\ldots$

$$c_1 = aenc_{k_1}(c_1 + \boxed{\text{meta data}})$$

$$\xrightarrow{c_1, \boldsymbol{t_1}}$$

tag	key
…	…

ck_0

$ck_1 \quad k_1 \quad \boldsymbol{t_1}$

$ck_2 \quad k_2 \quad \boldsymbol{t_2}$

$ck_3 \quad k_3 \quad \boldsymbol{t_3}$

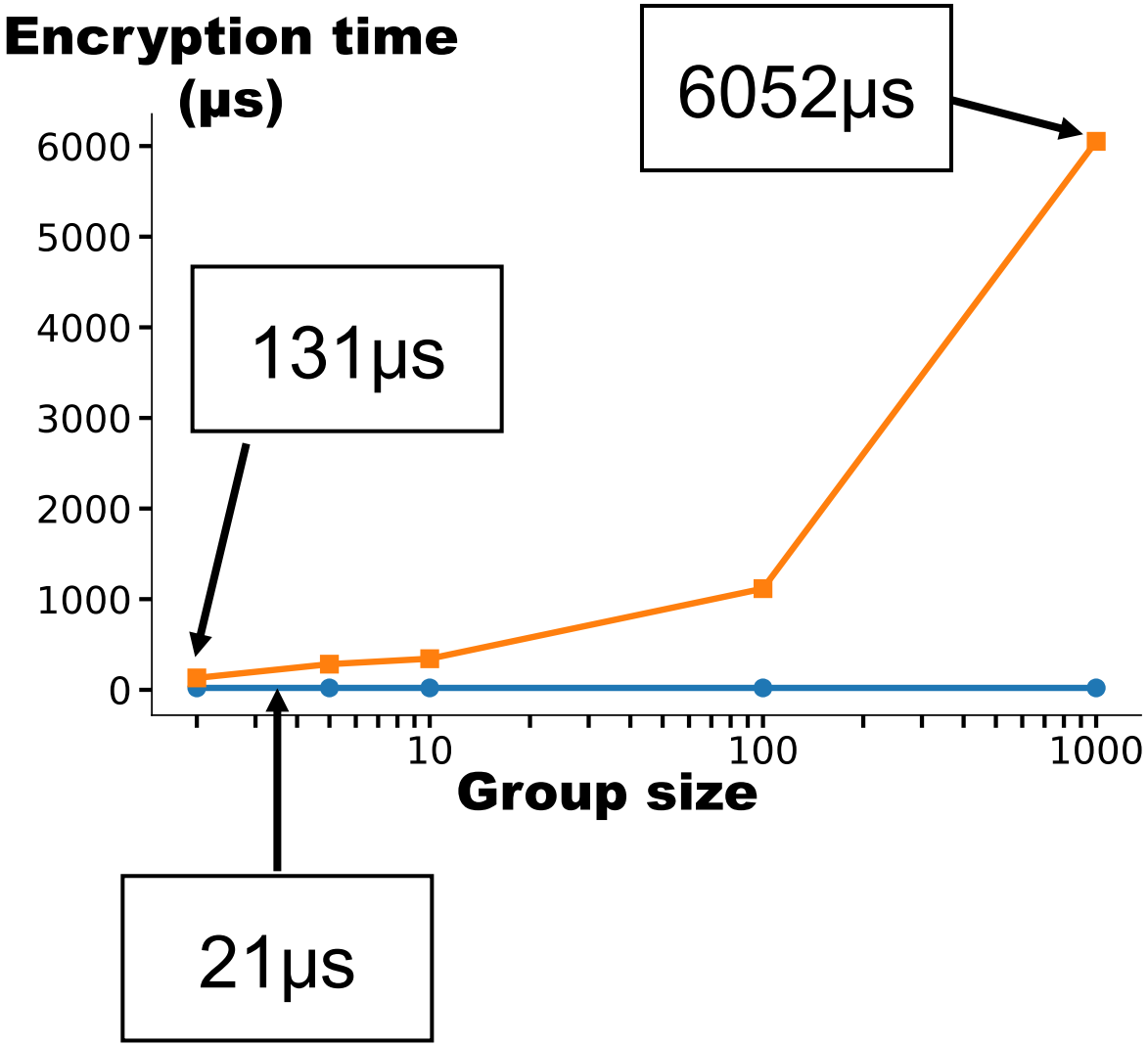
$\ldots$

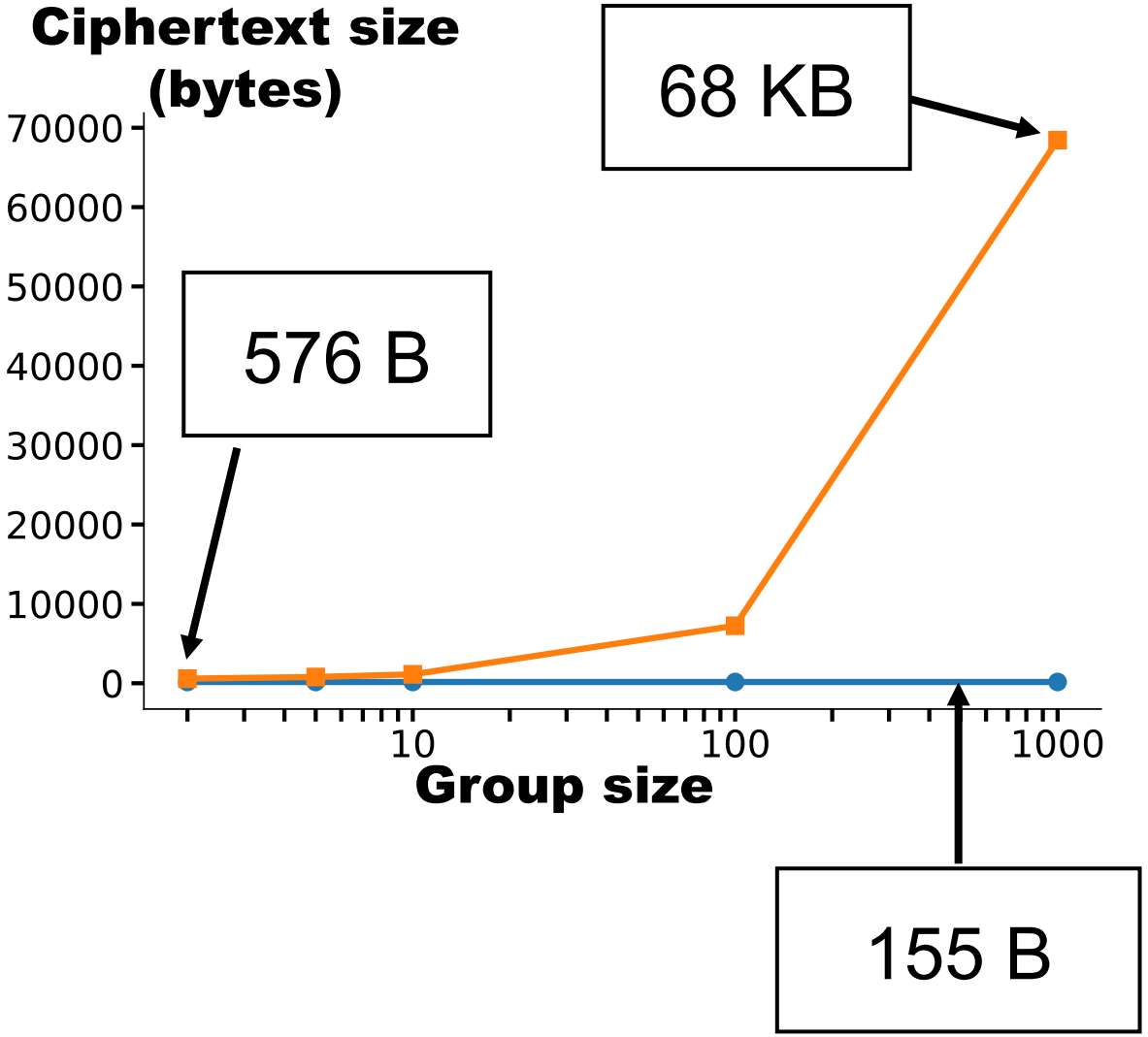
ck_0^*

$\ldots$

$$c_1 + \boxed{\text{meta data}} = adec_{k_1}(c_1)$$

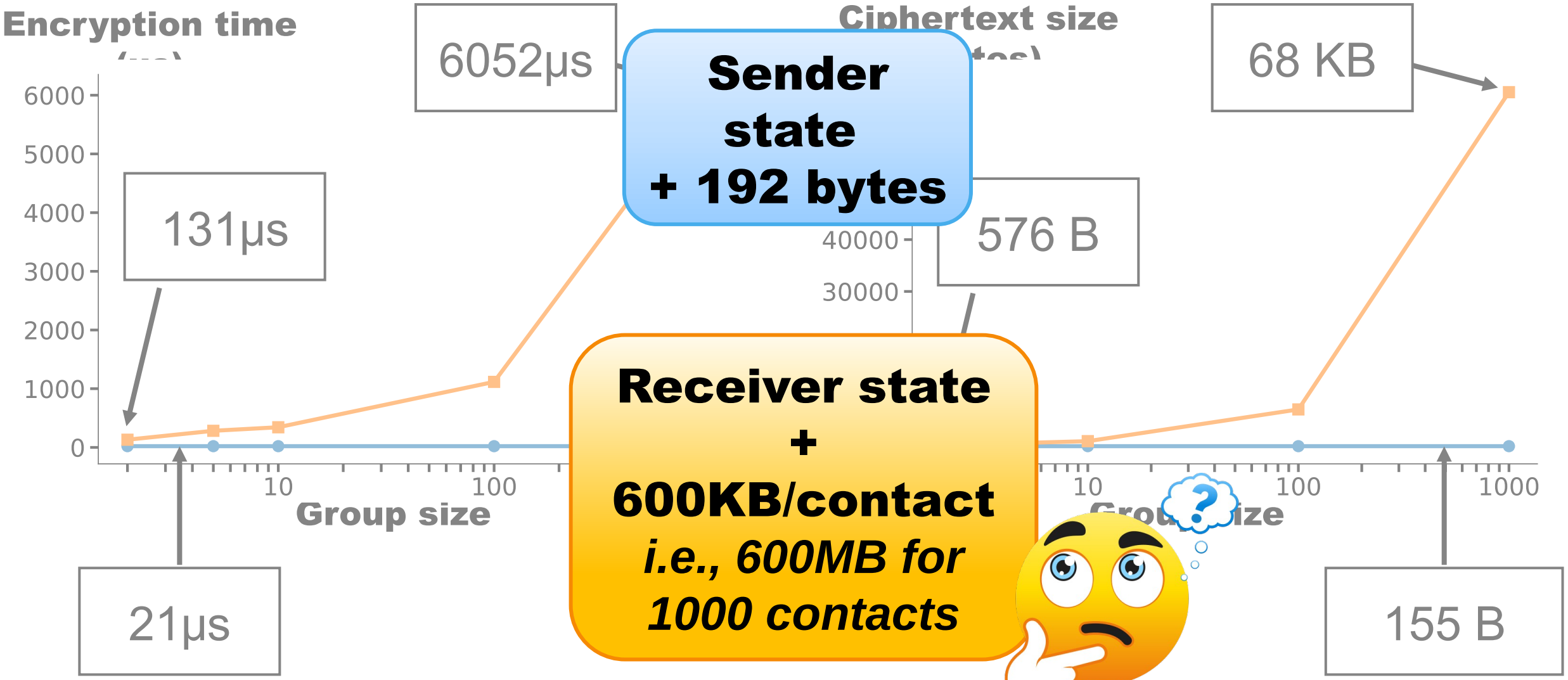
Anonymity Wrapper vs. Signal's Approach





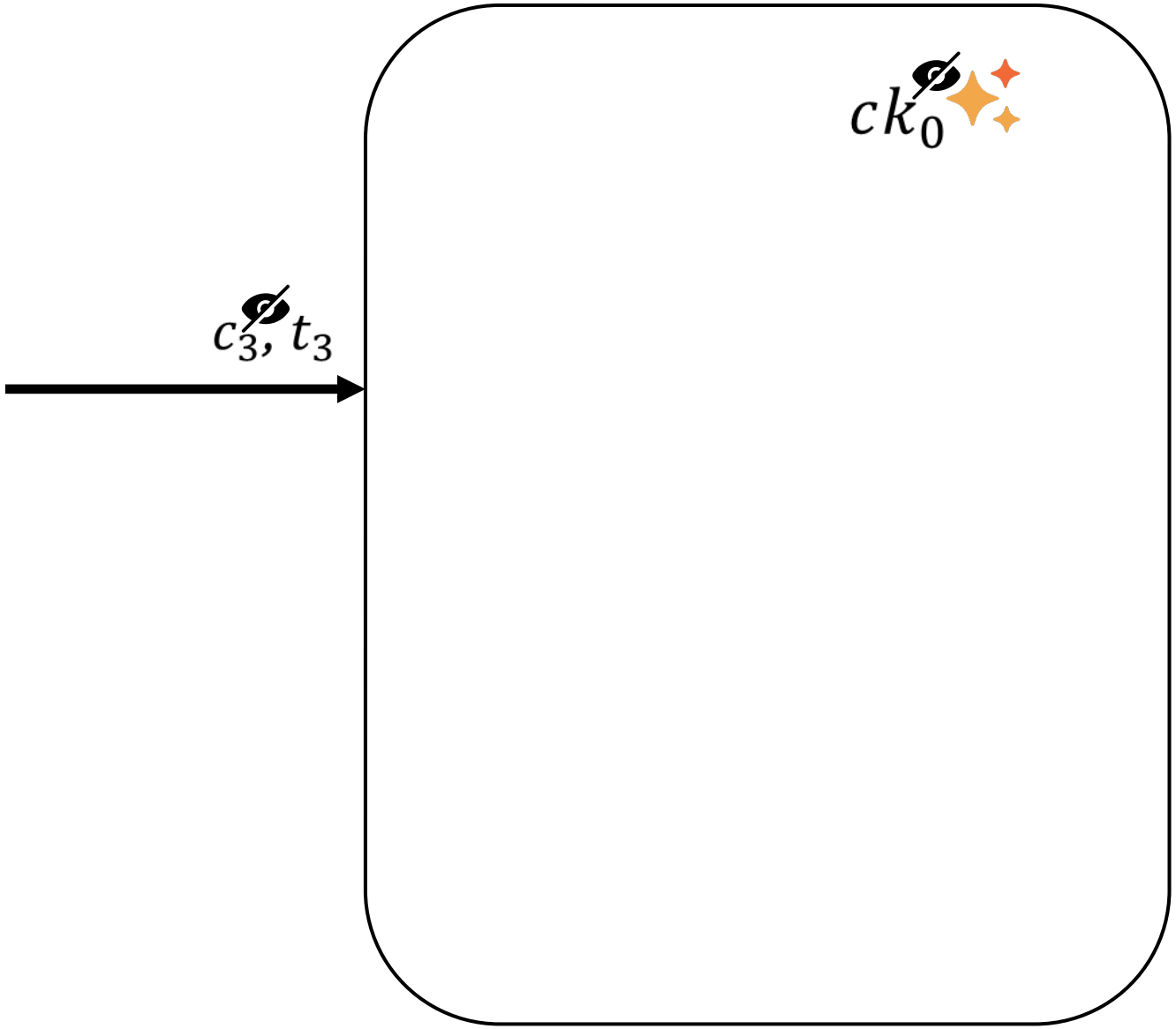
Experiments performed on 32 core Intel® Core™ i9-14900K

Anonymity Wrapper vs. Signal's Approach



Experiments performed on 32 core Intel® Core [illegible]-14900K

Why the Large Receiver State?



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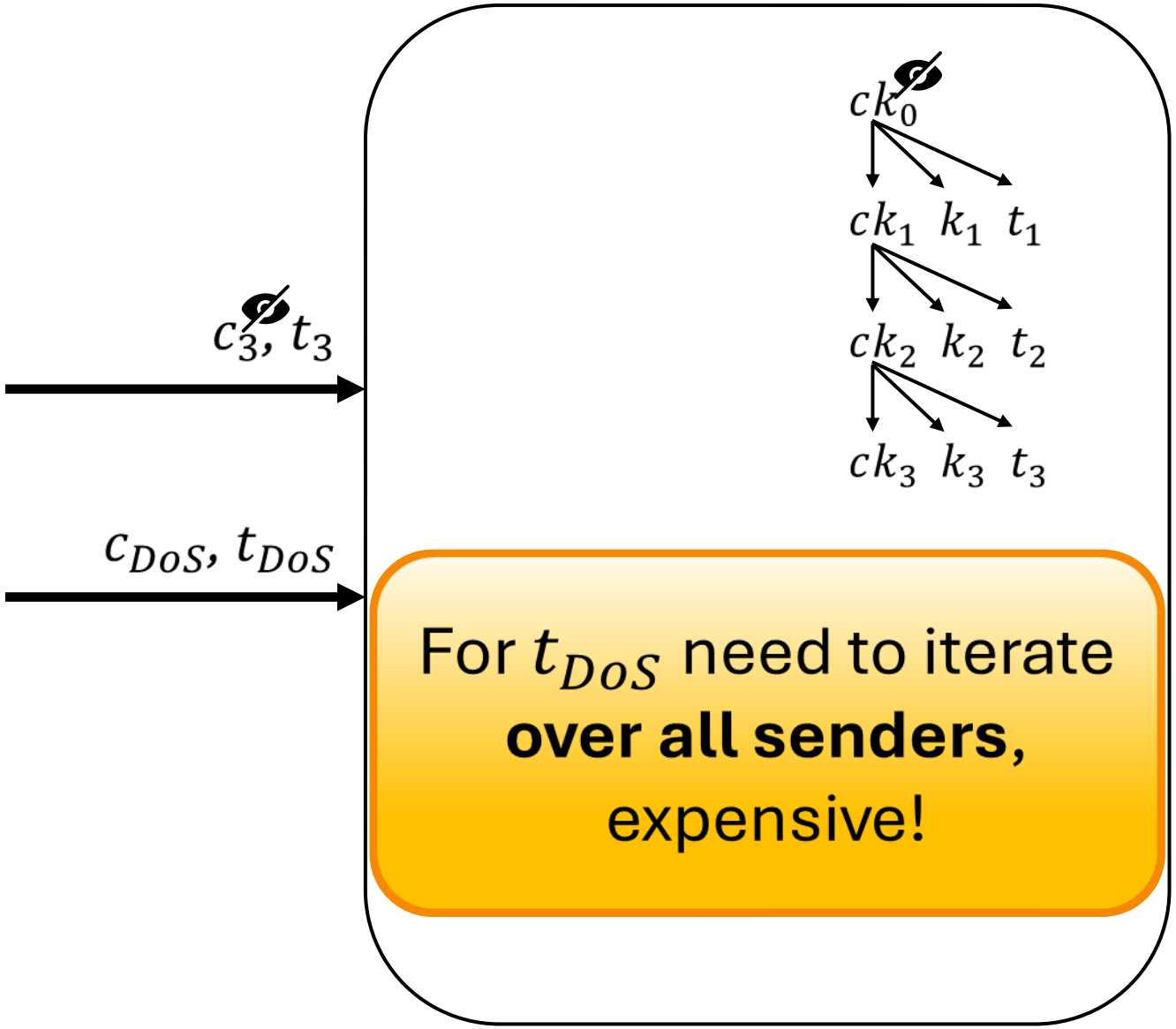
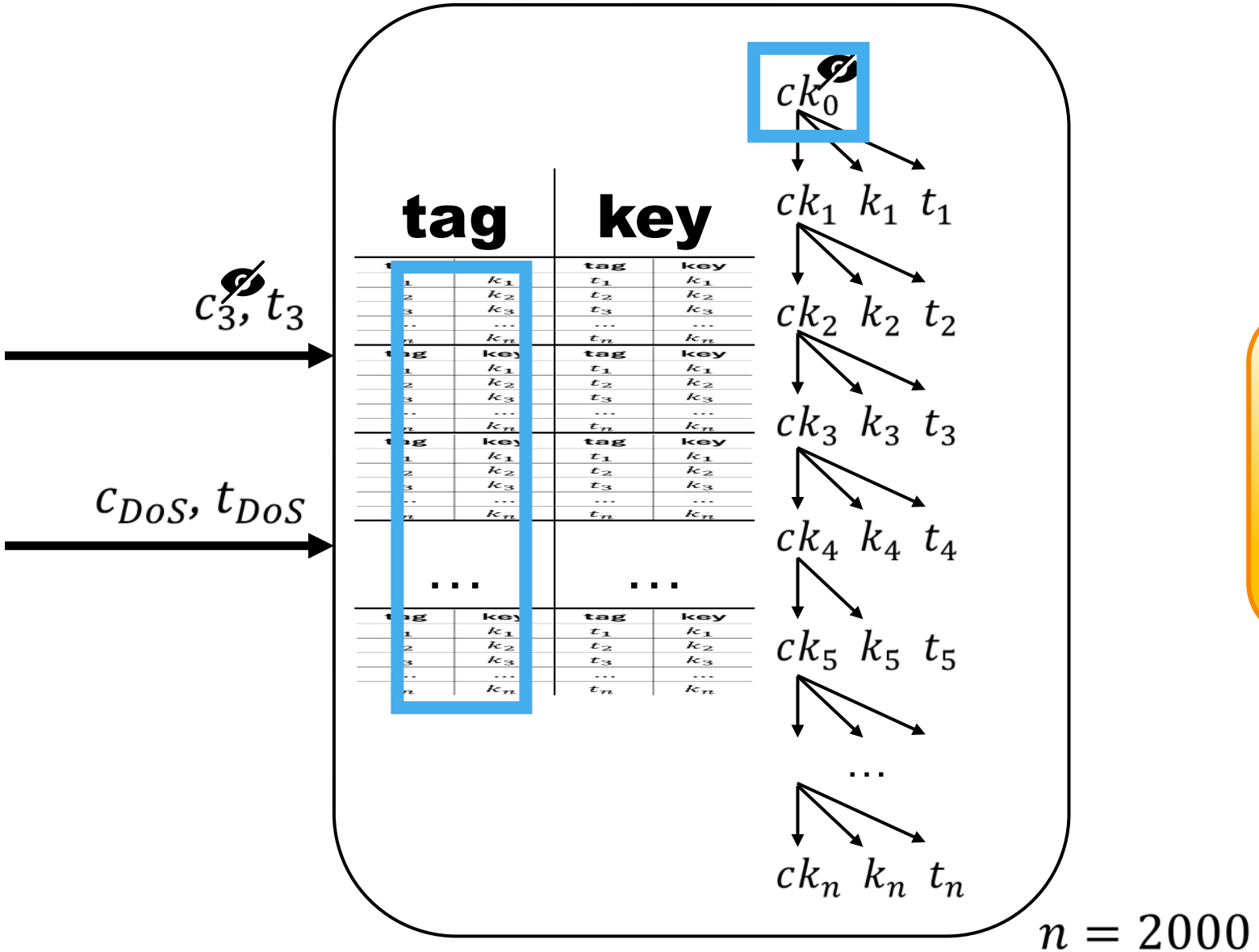


trial derive keys?

Why the Large Receiver State?

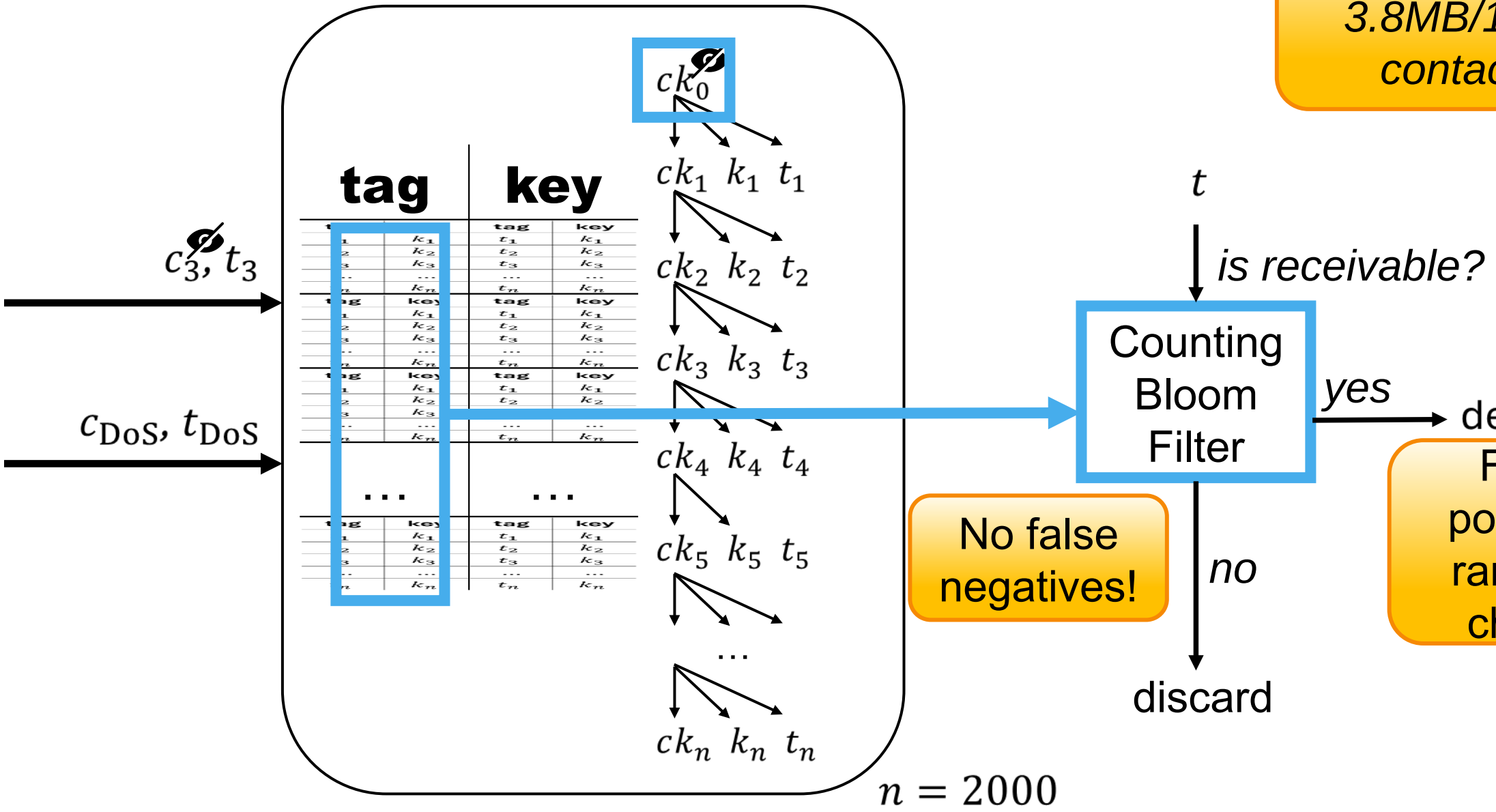


$n = 2000$

Pre-computed lookup table for each sender!

Fixing the Large Receiver State

+ 38KB/contact
3.8MB/1000 contacts



No false negatives!

False positives rare and cheap!

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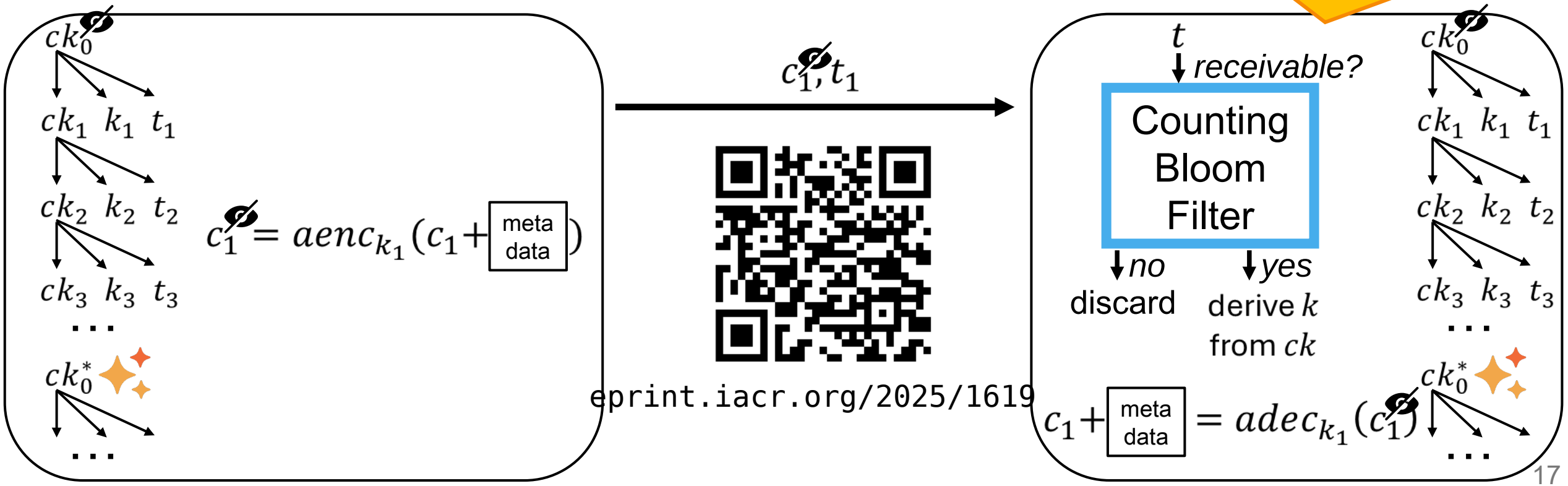
- Constant communication overhead
- Strong guarantees on temp. state exposure
- Signal considers deploying this protocol

Our group is looking for a Postdoc! ☺



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